Seeking help, she contacted each of the 29 scholars named on the journal's masthead as editors. She discovered that "many of them had no idea their names were being used." Others had been trying to get OMICS to remove their names, to no avail. Finally, the journal withdrew her paper, but Regan was infuriated. She believes that OMICS takes advantage of inexperienced scientists who are eager to be published. "They are really targeting grad students and scholars in the developing world," she says. "It's shameful." OMICS did not respond to repeated emails from *Science*.

The Federal Trade Commission is "working towards favoring some subscription based journal publishers who are earning millions of dollars from scientists."
OMICS Group Inc.

At stake in the case is "the integrity of the marketplace of ideas," says Gregory Ashe, an FTC attorney in Washington, D.C. In its 25 August complaint, the agency lays out a long list of charges, among them that OMICS used researchers' names to promote journals and conferences without their permission; falsely claimed journals are indexed by PubMed, the citations database maintained by the National Library of Medicine in Bethesda, Maryland; and untruthfully stated that its journals were granted independently calculated impact factor scores (a measure of citation rates). The agency wants the judge to order OMICS to end such practices, and also to consider forcing the company to refund "ill-gotten monies." (Bank records obtained by the agency show OMICS earned revenues of $26 million between 2009 and 2015.)

In a 22 September response, OMICS claims that FTC has no jurisdiction over its activities and that the case is a conspiracy on the part of its competitors to shut down open-access publishing. The agency is "working towards favoring some subscription based journal publishers who are earning billions of dollars from scientists," OMICS alleges on its website.

As *Science* went to press, FTC was waiting for the court to decide whether it will grant a preliminary injunction against OMICS, ordering it to stop deceptive practices until a final resolution of the case. ■



The first Polynesians crossed thousands of kilometers of open ocean in outrigger canoes.

PHOTOS: STEPHEN ALVAREZ/NATIONAL GEOGRAPHIC CREATIVE

HUMAN EVOLUTION

First Polynesians launched from East Asia to settle Pacific

"Game-changing" study of ancient genomes traces Polynesian roots solely to East Asian farmers

By **Ann Gibbons**

It was only 3000 years ago that humans first set foot on Fiji and other isolated islands of the Pacific, having sailed across thousands of kilometers of ocean. Yet the identity of these intrepid seafarers has been lost to time. They left a trail of distinctive red pottery but few other clues, and scientists have confronted two different scenarios: The explorers were either farmers who sailed directly from mainland East Asia to the remote islands, or people who mixed with hunter-gatherers they met along the way in Melanesia, including those from Papua New Guinea.



The Lapita people left behind a trail of distinctive stamped red pots at about 100 archaeological sites in Oceania.

Now, the first genome-wide study of ancient DNA from prehistoric Polynesians has boosted the first idea: that these ancient mariners were East Asians who swept out into the Pacific. It wasn't until much later that Melanesians, probably men, ventured out into Oceania and mixed with the Polynesians, according to the new study.

"The paper is a game-changer," says Cristian Capelli, a population geneticist at the University of Oxford in the United Kingdom, noting that it settles a decades-long dispute. By showing that the East Asians hopscotched past islands already populated by Melanesians without picking up their genes, it is also a case study in how culture can initially bar mixing between groups. "Farmers move in and don't mix much with the hunter-gatherers," says evolutionary geneticist Mark Thomas of University College London. "We see this again and again and again" elsewhere in the world.

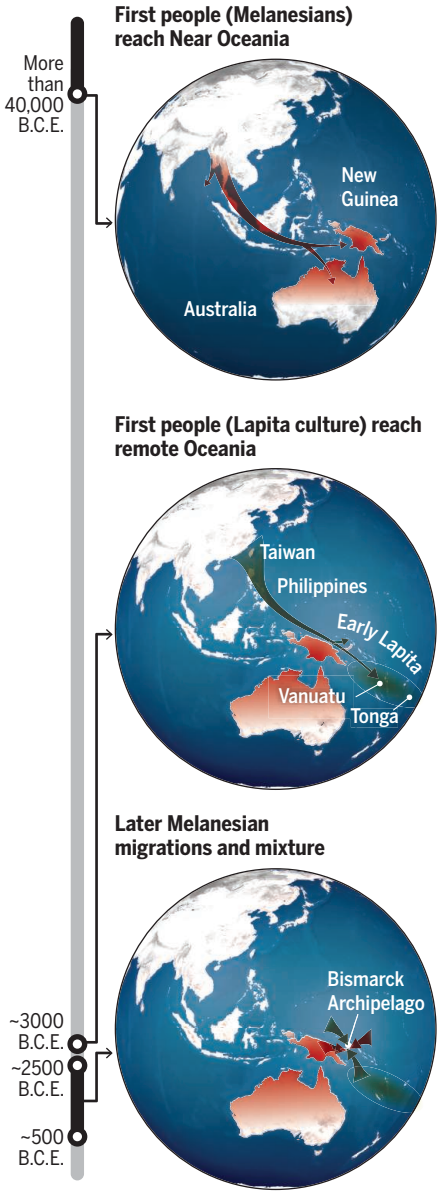
The first Polynesians left plenty of tantalizing artifacts, including their stamped red pottery, obsidian tools, and shell ornaments. Collectively known as the Lapita culture, this set of artifacts first appeared more than 3000

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Peopling the Pacific

At least 3000 years ago, the first Polynesians sailed past New Guinea without mingling with the Melanesians already living there, and settled the distant islands of Vanuatu, Tonga, and beyond.



MAPS: V. ALTOUNIAN/*SCIENCE* ADAPTED FROM SKOGLUND *ET AL.*/*NATURE* (2016)

years ago in the Bismarck Archipelago east of New Guinea (see map, above). The Lapita grew taro, yams, and breadfruit, and brought pigs and chickens with them as they spread rapidly to the islands of Vanuatu and New Caledonia and eventually to Fiji, Tonga, Samoa, and beyond.

Back in 1985, archaeologist Peter Bellwood of the Australian National University in Canberra proposed that the Lapita had roots in farming cultures in East Asia. Based on dating of Lapita sites, he proposed that they moved rapidly from mainland China to Taiwan and the Philippines, then out across the open ocean from Vanuatu to Samoa, covering 2300 kilometers in about 300 years. This "express train" picture fit with linguists' models, in which Austronesian languages spread from East Asia into Oceania and were distinct from Papuan languages in Melanesia.

But other researchers argued that the DNA of living Polynesians showed evidence that their Lapita ancestors had lingered in Melanesia, mixing with the locals and slowly spreading eastward. This so-called "slow boat" model had prevailed in recent years (*Science*, 2 March 2001, p. 1735).

In the new study, an international team extracted ancient DNA from four skeletons, all female, from the islands of Vanuatu and Tonga and dated from 2300 to 3100 years ago; three skeletons were directly associated with the Lapita culture. The team sequenced the DNA at up to 231,000 positions across the genomes of each woman, and compared the sequences with those of nearly 800 present-day people from 83 populations in East Asia and Oceania.

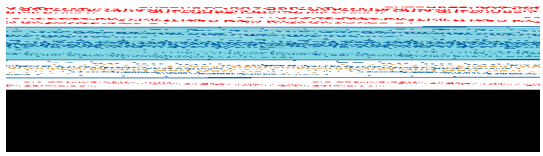
The four women were from a distinct population that had no evidence of mixing with the ancestors of people living in Papua New Guinea today, as the team reports in *Nature* this week. Instead, the women shared all their ancestry with the indigenous Atayal people in Taiwan and the Kankanaey people in the Philippines. "The Lapita have no evidence for Papuan ancestry," says co-author Pontus Skoglund, a postdoc in David Reich's lab at Harvard Medical School in Boston. That suggests that their ancestors rode the fast train, sweeping all the way to Oceania without mixing with Melanesians on the way.

Polynesians today do carry a significant amount of Melanesian DNA. But that DNA is in relatively long, unbroken chunks, the analyses found, suggesting that it was incorporated into Polynesians' genomes recently, after the Lapita period, perhaps about 500 to 2500 years ago. The team also noted that the Polynesian X chromosome has less Melanesian DNA than other nuclear chromosomes. Because X chromosomes are more likely to be inherited from mothers (sons get only a Y from their fathers), that suggests that much of the Melanesian DNA came through the male line, as Melanesian men interbred with Polynesian women. "The female ancestors of modern-day Oceanians are mainly Lapita, whereas their male ancestors include Papuans," Skoglund says.

Bellwood is pleased. "All the evidence from different disciplines has come together," he says. By getting DNA from the ancient Polynesians themselves, "the genomics has clinched it." ■

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Editor's Summary

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